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ROLL NO:43

PROBLEM STATEMENT: In an e-commerce system, customer account IDs are stored in a list, and you are tasked with

writing a program that implements the following:

- Linear Search: Check if a particular customer account ID exists in the list.
- Binary Search: Implement Binary search to find if a customer account ID exists, improving the search efficiency over the basic linear.

'''

```
emp_id=[]
```

```
n=int(input("Enter the size of the array:"))
```

```
for i in range(n):
```

```
    emp=int(input("Enter employee id:"))
```

```
    emp_id.append(emp)
```

```
def linear_search(emp_id,target):
```

```
    for i in range(0,len(emp_id)):
```

```
        if(emp_id[i]==target):
```

```
            return i;
```

```
            break;
```

```
    return-1
```

```
def binary_search(emp_id,key):
```

```
    low=0
```

```
    high=n-1
```

```
while low<=high:
    mid=(low+high)//2
    if emp_id[mid]==key:
        return mid;
    if emp_id[mid]<key:
        low=mid+1
    else:
        high=mid-1
return -1
```

```
def menu():
    print("1.linear:")
    print("2.binary:")
    print("3.Exit:")
```

```
while True:
    menu()
    choice=input("Enter your choice(1-3):")
    if choice=='1':
        key=int(input("Enter id to be search:"))
        result=linear_search(emp_id,key)
        if result!=-1:
            print("Element found at location",result)
        else:
            print("Employee Id is not found")
    elif choice=='2':
```

```
sorted_id=sorted(emp_id)
key=int(input("Enter id to be search:"))
result=binary_search(sorted_id,key)
if result!=-1:
    print("Element found",result)
else:
    print("Element not found",result)
elif choice=='3':
    print("Exit")
    break;
else:
    print("Invalid Choice")
    Enter the size of the array:3
```

'''Output:

gescoe@gescoe-OptiPlex-3020:~/SE divB \$ python3 linear search.py

Enter the size of the array:3

Enter employee id:45

Enter employee id:67

Enter employee id:89

1.linear:

2.binary:

3.Exit:

Enter your choice(1-3):1

Enter id to be search:23

Employee Id is not found

1.linear:

2.binary:

3.Exit:

Enter your choice(1-3):2

Enter id to be search:45

Element found 0

1.linear:

2.binary:

3.Exit:

Enter your choice(1-3):3

Exit

'''